

Custom SSRF Detection Agent, Based on IDS-Agent

Draft for feedback · Code: github.com/blastre/ml-web-security-gateway

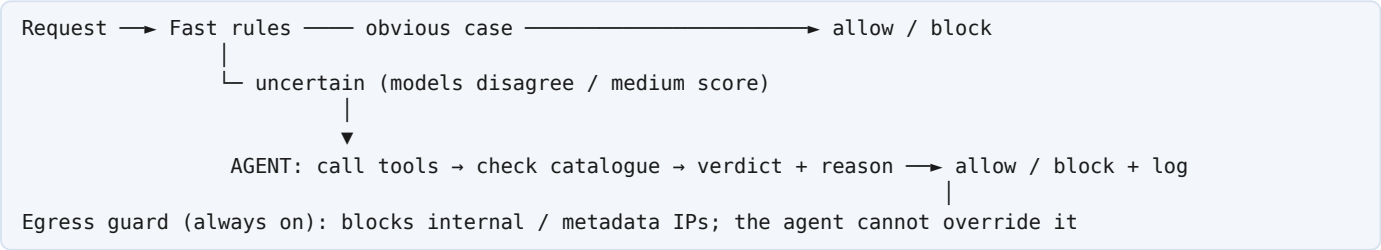
Idea. Replace Codex, which only explained incidents after they were blocked, with our own agent. It checks incoming requests on the VM and decides: **is this SSRF, which technique, and why**.

Base paper

IDS-Agent: An LLM Agent for Explainable Intrusion Detection (Li, Xiang, Bastian, Song, Li; NeurIPS 2024 Workshop). An LLM agent that reasons step by step, **calls ML classifiers as tools**, looks things up in a **knowledge base**, and returns a **verdict with an explanation**. It beat single models and majority voting, especially when the classifiers disagreed and on unseen attacks.

How we adapt it to SSRF

IDS-Agent	Our agent
Input: IoT network traffic	Input: the request URL plus where it really goes (resolved IP, redirects)
Tools: ML classifiers	Tools: our 3 models (Logistic Regression, Random Forest, Isolation Forest) and URL rules
Knowledge base	Our catalogue of 22 SSRF techniques and the OWASP SSRF guidance
Output: attack or benign, plus reason	Output: SSRF or benign, technique, reason



What is new

- IDS-Agent was tested on IoT network traffic. We apply it to **SSRF, a web attack**, detected **live on a real VM**.
- Existing AI work on SSRF either **classifies URLs** (LSTM, GRU, XGBoost) or **scans source code** (Artemis, SSRFSeek). None uses an agent on live requests, and none checks the final destination.

References

- Y. Li et al., *IDS-Agent: An LLM Agent for Explainable Intrusion Detection in IoT Networks*, NeurIPS 2024 Workshop.
- B. Kneip et al., *Sample-Efficient LLM-Based Detection of Malicious Web Server Logs* (CSIC 2010), arXiv, 2026.
- Artemis: Toward Accurate Detection of Server-Side Request Forgeries through LLM-Assisted Taint Analysis*, Proc. ACM Program. Lang., 2025.
- Q. Lan et al., *Silent Egress: When Implicit Prompt Injection Makes LLM Agents Leak Without a Trace*, arXiv, 2026.
- Benchmarking Deep Learning Models for Detecting SSRF Vulnerabilities: A Comparative Study*, IEEE, 2024.